

# 5

## THE DYNAMICS OF CONTRAST

The strongest control of visual effect lies in the understanding of the connection between message and meaning and the visual techniques. The syntactical guidelines offered by the psychology of perception, the familiarity with the character and appropriateness of the essential visual elements, provide those seeking visual literacy a firm foundation for compositional decisions. But the crucial control of visual meaning lies in the focusing function of the techniques. And of all the many visual techniques we will be investigating, none is more important to the control of a visual message than contrast.

### CONTRAST AND HARMONY

Visual techniques, as already noted, have been arranged in polarities not only to demonstrate and accentuate the wide range of operative choices available in the design and interpretation of any visual statement, but, also, to express the intense importance of the technique and concept of contrast to the entire medium of visual expression.

All meaning exists in the context of polarities. Would there be understanding of hot without cold, high without low, sweet without sour? The contrast of substances and the sensorium's responsiveness to it dramatizes meaning through the opposite formulations. "The basic principle of 'form' determines that close relation between apperceptive unity and logical distinctions which was known to the ancients as 'unity in diversity.'" This is how Susanne Langer, in her essay, "Abstraction in Science and Abstraction in Art"\* describes the "articulation of structural elements of a given whole." In the process of visual articulation, contrast is a vital force in creating a coherent whole. In all of art, contrast is a powerful tool of expression, the means for intensification of meaning and, therefore, of the simplification of communication.

Although on the list of techniques harmony is advanced as the polarity of contrast, it should be noted with great emphasis that the importance of both have a deeper significance in the entire visual process. They represent an ongoing and highly active process in how we see visual data and, consequently, how we understand what we see. The human organism seems to seek harmony, a state of ease, of resolution, what the Zen Buddhists speak of as "meditation in supreme repose." There is a need to organize all stimuli into rational wholes

\*In Problems of Art



as demonstrated by the experiments of the Gestaltists. To reduce tension, to rationalize and explain, to resolve confusion, all appear dominant in man's needs. Only in the context of the logical conclusion of this endless and active quest does the value of contrast become clear. If the human mind got what it so fervently sought in all of its thought processes, what would it be? A state of weightless, fixed, motionless balance—absolute balance. Contrast is a counterforce to this human appetite. It unbalances, shocks, stimulates, arrests attention. Without it, the mind would move toward eradicating all sensation, creating a climate of death, of nonbeing. Whether or not we all feel a strong death wish, a nagging little voice in the ear of the trapeze artist whispering "let go," the state of absolute resolution, closure, is not final or finished or zero sensation enough. Like the all middle-gray environment, there would still be a sensation of sight without seeing, life without living. We would be like Palinurus, buried alive and cursed to sense all things from the grave, a living death. Psychologists tell us that our dreams are a kind of perspiration of the mind, throwing off the poisons of the psyche in a constant process of cleansing and clarification and absolutely essential to our mental health. So, also, the very process of living seems to demand a richness of sensory experience, especially through sight. We see so much more than we need to see, but our visual appetite is never satisfied. We reach out to the world and its complexities through our vision and we reach out through what the poets call "our mind's eye" to think visually. If, in its workings, the visual process moves toward absolute neutrality, it is the process and not the end result that should concern us.

### THE ROLE OF CONTRAST IN SEEING

In visual literacy the importance of the meaning of contrast begins at the base level of seeing or not seeing at all through the presence or absence of light. No matter how well the physiological equipment of sight functions, the eyes, the nervous system, the brain, or how much there is in the environment to see, in a totally dark circumstance, we are all, in effect, blind. The human equipment for sight is of secondary importance; light is the key visual force. In its elemental visual state, light is tonal, reaching from brightness (or lightness) to darkness in a series of steps which can be described as having very subtle gradations. In the process of seeing, we are dependent on observing the interactive juxtaposition of those gradations of tone in order to see objects. Remember that the presence or absence of color does not affect tonal values; they are constant and hold infinitely more importance than color in seeing as well as designing and making. In pigment, lightness is synthesized or suggested by whiteness toward absolute white,

while darkness is suggested by blackness toward absolute black. So everything we see can be invested with both properties of tonal values, the pigment quality of relative whiteness or blackness of tone and the physical quality of lightness or darkness. Physical light has a broad range of tonal intensity, while pigment is usually used in a limited range of eight to fourteen tonal steps. The broadest range of distinctly different gray tones in pigment is about thirty-five. Without light upon it, the whitest of whites will not be seen. So, whether from the sun or the moon or candlelight or the incandescence of the electric light bulb, light is the essential link in our physiological capacity of vision.

But the absence of light is not unique in its potential for blocking sight. If our entire environment were composed of an equal and matching value of a middle tone of gray, halfway between black and white, it would be possible to see; that is, we would not be experiencing the sensation of blindness created by a totally black environment. However, the ability to discern what we were seeing would be entirely erased from our perceptions. In other words, the contrast of tone is as vitally important as the presence of light in the process of seeing. Through tone, we can perceive patterns that we simplify into objects with shape, dimension, and other elemental visual properties. It is a decoding process of constant simplification of the raw data until, through it, we come to recognize and learn about the world we live in from the ants, moving about busily on the ground, to the stars, flickering with various sizes and tonal intensities in the sky. Light creates patterns, and once these patterns are identified, the information gained is stored in the brain to be utilized in recognition. It is an intricate and tricky process, described artfully by Bernard Berenson in his essay, *Seeing and Knowing*: "I see masses of green, opaque or translucent or glistening. They are spiky or smooth and, as if supporting them, roughly cylindrical somethings, vaguely brownish, greenish, grayish. I have learned in babyhood that they are trees and I vest them with trunks, with branches, with twigs, with foliage, with single leaves, according to their presumed species, ilex, chestnut, pine, olive, although my eyes see only varying shades of green."

So the eyes and the process of sight extend in many directions beyond seeing into the realm and workings of intelligence. The entire nervous system interacts with sight, enhancing our ability to discriminate. Touch, taste, hearing, smelling all contribute to our comprehension of the world around us, augmenting and, sometimes, contradicting what our eyes tell us. We touch something to determine if it is soft or hard; we smell something to discover if it is fragrant; we taste



something to find if its pleasant odor indicates that it is equally pleasant to eat; and we listen to know if something is moving or still. All our senses are discriminating and constantly refining our recognition and understanding of the environment. But, of all our senses, there is little argument that the one we depend on the most, the one that has superior power, is sight. And sight functions most effectively when the patterns we observe are visually clarified through contrast. In nature and in art, contrast is of key importance to the visualizer in what Donald Anderson in his book, *Elements of Design*, calls "manipulating a set of such raw materials as clay, wire, pigment, data, sounds, words, numbers ... transforming them into cohesive structures on a higher level of significance."

### THE ROLE OF CONTRAST IN COMPOSITION

Vision is intensely involved in pattern perception, a process that legislates toward a need for discernment. R. L. Gregory in *The Intelligent Eye* says: "In this sense 'patterns' are very different from 'objects.' By a pattern we mean some set of inputs, in space or time, at the receptor." Seeing means sorting out the patterns toward the end of understanding or recognition. Ambiguity is its natural enemy and must be avoided if the process of seeing is to function properly. We observe a tree. If it is upright and solid looking, we know we can lean on it. If it looks dangerously tipped and loose, we would not trust our weight to it. But if it is a little bit of both, not quite shaky, but on the other hand, not strong and upright either, then we are faced with confusing visual information. The pattern, the visual input, is inconclusive. Other methods of testing the sturdiness of the tree would have to be used. A line drawn through a square very close to dead center and yet off of it, is a more abstract example of the same con-

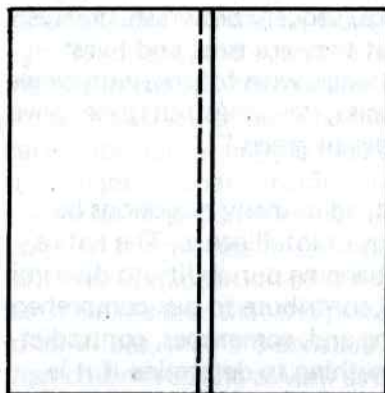


FIGURE 5.1

dition (5.1). The line is just enough off of the felt axis to disturb and upset the viewer, but not enough off to make its unbalanced position distinctly clear. The most effective utilization of the mechanisms of visual perception is to position or identify visual clues as one thing or the other, in balance or off, strong or threateningly weak. The Gestaltists deal with this need by describing the two opposite visual states as leveled or sharpened. Koffka, in his book, *Principles of Gestalt Psychology*, defines sharpening as "an increase or exaggeration" and leveling as "a weakening or toning down of a peculiarity of a pattern." In the terminology of visual techniques, sharpening can be equated with contrast (5.2), while leveling can be associated with harmony (5.3). But whatever descriptive language is used to describe the two polarities of visual composition, leveled or sharpened, the point must be emphasized that they are both excellent tools for constructing a visual statement with clarity of point of view. The use of them in a craftsmanlike manner goes a long way toward avoiding confusion for both the designer and his viewer.

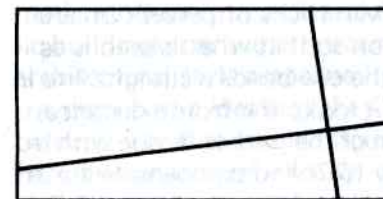


FIGURE 5.2

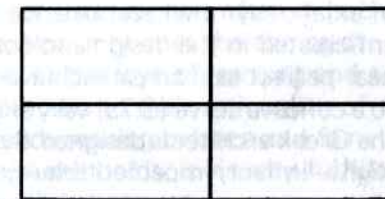


FIGURE 5.3

What the Gestaltists have researched and established through their recognition of the value of these two visual techniques is that the eye (and along with it, the human brain) will not be deterred in its endless pursuit of resolution or closure in the sensory data it views. Wertheimer introduced the principle which governs this hypothesis and called it "the law of Prägnanz," which he defines as follows: "psychological organization will always be as 'good' as the prevailing conditions allow." What is meant by "good" is not entirely clear. No doubt what he is suggesting is resolution in terms of regularity, symmetry, simplicity. Such forces as the need to finish off or connect an unfinished line (5.4) as in closure or to match like shapes as in the "principle of similarity" apply here (5.5). Finishing the lines or grouping the similar shapes is a step toward simplification, an inevitable step in the mechanics of perception in the human organism. But, is it as desirable as the physiological pull toward it would indicate? Absolute regularity can be refined and regulated toward a perfect end result in a visu-



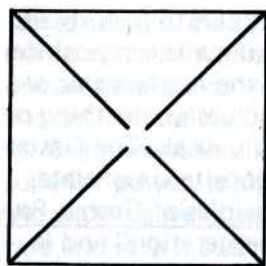


FIGURE 5.4

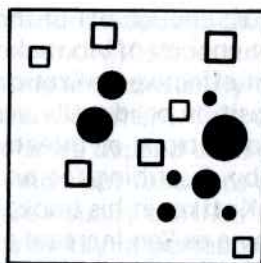


FIGURE 5.5

al statement. It is easy to determine and simple to respond to. Nothing is left to chance, or emotion, or subjective interpretation at either end of the stimulus  $\longleftrightarrow$  response communication model. The Greeks demonstrate the absolute and logical pursuit of harmonious results in the design of such temples as the Parthenon. Not only is there a utilization of the formula of the "Golden Mean," the mathematically determined proportion, but there is the most complete use of axial or symmetrical balance (5.6). Even tricks of perception are anticipated in the design and construction so that what is seen is as near perfect as man can achieve. Since the eye bends a straight line into a concave curve (5.7a) very slightly as it looks at it from a distance, the Greek architects designed the column of the temple facade with a slight—in fact, imperceptible—convexity (5.7b) to compensate for the phenomenon and to produce an apparently perfect straight line (5.7c). They would stop at nothing in their pursuit of perfection. The final effect was what they really sought, an effect that was completely har-

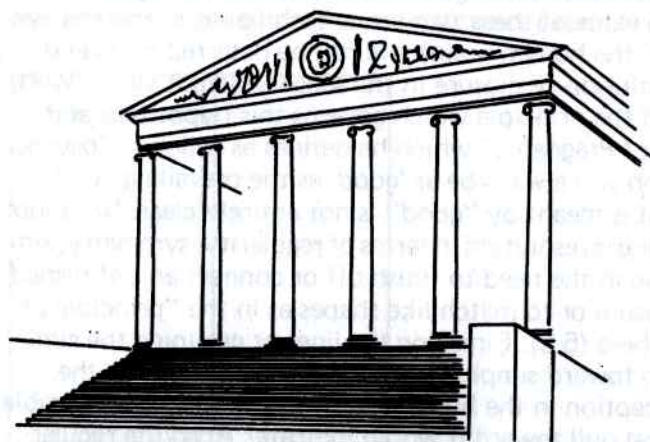


FIGURE 5.6

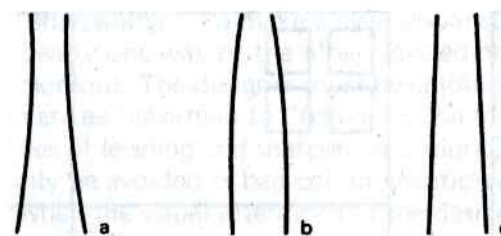


FIGURE 5.7

monious and balanced, in which nothing was left visually unresolved. We call the Greek style "classical," and associate with the quality of that style a total stability, with no equivocation on the part of the designer and no factors to unsettle the viewer. Certainly it all answers to the criteria that will produce the ideal of Wertheimer's "good" described in his "law of Prägnanz"; it conforms to the unconscious demands of the mind and the physical mechanics of the body. It is a quality that official institutions have certainly appropriated in the modern, Western world, and the classical style appears often in public buildings, particularly courthouses. Not only does the choice of architectural style associate its builders to the love of learning and the democratic ideals of the Greeks, but also to their rationality of balance. The blindfolded figure of Justice seeking balance and fairness, as shown symbolically in the scales she holds, is visually fulfilled in the symmetry of the design of a Greek temple.

But "good" as it is defined in the "law of Prägnanz" does not need symmetry and balance as its only expression; "good" in this sense also describes the clarity in a visual statement that can be produced through sharpening, or, as it can also be defined, the technique of contrast. Even though the obvious and most apparent need of the human being is for balance and repose, the need for resolution is equally strong, and sharpening offers great potential for achieving it, resolution that is an extension of the inner idea of harmony and is drawn from the organization of complexity rather than pure simplicity. Rudolf Arnheim in *Art and Visual Perception* calls the apparent contradiction in this fact "a duality connected to the parallel activities of the process of growth and the striving for vital aims." Leveling (5.8), as in the design of the facade of a Greek temple, is harmonious and simple, but sharpening (5.9) has much more vital intentions in its visual character. And yet it would be unfair to say that one was easier to perceive than the other. They are merely different.



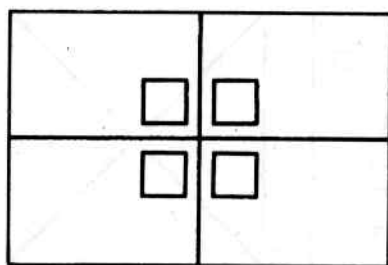


FIGURE 5.8

The act of seeing is a process of discernment and judgment. In Figure 5.8, both processes can be activated and the results of their workings can be established quickly and automatically by the viewer. The example demonstrates complete and indisputable balance. But the same quick and automatic response on the part of the viewer to Figure 5.9 can be predicted. The definition of the structure is not as unequivocal except in a negative sense; the visual elements are not symmetrical. It does not balance in the obvious sense that the elements in Figure 5.8 do. But balance does not have to take the form of symmetry. The weight of the design elements can be adjusted asymmetrically. The additional forces move the design away from simplicity, but the final effect is one of balance structured by weight and counterweight, action and counteraction. The final effect can be read and responded to quite clearly by the viewer; it is just a more complex process and therefore slower (5.10). The same psychophysiological human perceptual ability that establishes symmetrical balance can automatically measure and respond to asymmetrical balance. The process is not as easily demonstrated and defined, and, consequently, often seems intuitive rather than physical.

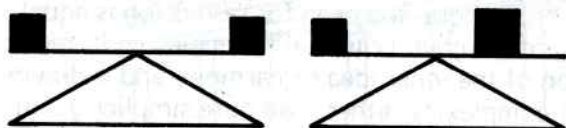


FIGURE 5.10

One thing is certain about the asymmetrical balance of Figure 5.10, and that is that it is not almost balanced symmetrically. The viewer is not provoked by lack of resolution or upset with visual ambiguity. The design is clearly nonaxial balance, and because of the clarity of this fact we can say it is a good demonstration of the state of visual

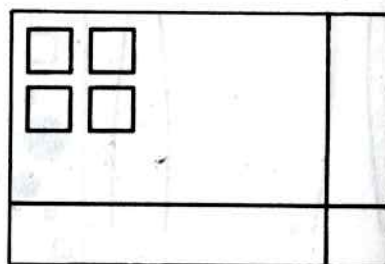


FIGURE 5.9

"sharpening." To make a clear visual statement, the choice must go clearly one way or the other, leveled or sharpened, contrasted or harmonious. The designer must be enjoined to follow the dictum of the Yankee fisherman to "fish or cut bait." The area between the polarities of leveling and sharpening is murky and unclear, and should usually be avoided as bad communication and as aesthetically ugly. When the visual intentions of the designer are not strongly delineated and controlled, the result is ambiguous and the effect created is an unsatisfying and frustrating one for the audience (5.11). The balance cannot be established clearly one way or the other; the elements cannot be organized and related, primarily to one another, and, secondarily, to the field. Unless that is the visual impression that is sought by the designer (an unlikely possibility), ambiguity should be avoided as the most undesirable visual effect, not just because it is psychologically upsetting, but because it is sloppy and inferior on any level of the criteria of visual communication.

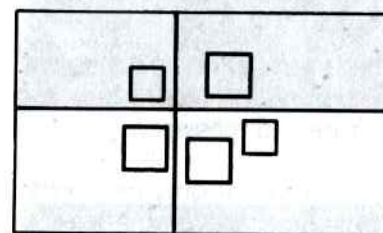


FIGURE 5.11

Harmony, or the leveled state in visual design, is a useful and almost foolproof method for the solution of compositional problems for the inexperienced and unskilled visual message-maker. The rules to follow are as simple and clear as can be, and if they are followed rigidly, the results are predictably attractive. You just cannot go wrong. For reasons of insurance and assurance, axial balance as a design strategy has been an invaluable aid in the constructing of clean, neat designs. Book design has been dominated by the classical look of pages in absolute balance (5.12), especially since the invention of metal movable type. The mechanical, mathematical nature of typesetting lends itself perfectly to formula computations that produce balance. But, as safe and sure as the harmonious technique offered by leveled design may be in providing, as it does in book design, a nonintrusive visual setting for the message, the mind and the eye demand stimulation. Boredom is as much a threat in visual design as it is elsewhere in art and com-



## A PRIMER OF VISUAL LITERACY

Donis A. Dondis

FIGURE 5.12

munication. The mind and the eye demand stimulation and surprise, and an approach to design that functions boldly and successfully suggests the need for sharpening of the structure and message.

As a visual strategy to sharpen meaning, contrast not only can excite and attract the attention of the viewer, but it can also dramatize that meaning to make it more important, more dynamic. If, for instance, you want something to look clearly large, put something small next to it. This is contrast, an organization of the visual stimuli toward an intense effect. But intensification of meaning goes even further than juxtaposition of unlike elements. It consists of a deletion of the superficial and unnecessary, which leads to a natural spotlighting of the essential. Rembrandt followed this method in the development of his technique of "chiaroscuro." The name of the technique is drawn from the combination of two Italian words: chiaro, which means light, and scuro, for dark. That is what he uses, lightness and darkness. In his paintings (5.13) and in his etchings (5.14), Rembrandt discarded the middle tones to accent and highlight his subject matter with a striking, theatrical look. The incredible richness of result is as strong an argument for the understanding and utilization of contrast as can be found in the body of visual work at any level.

Contrast is an essential tool in the strategy of control of visual effects and, consequently, meaning. But contrast is simultaneously a tool, a technique, and a concept. In basic terms, we have a deeper under-

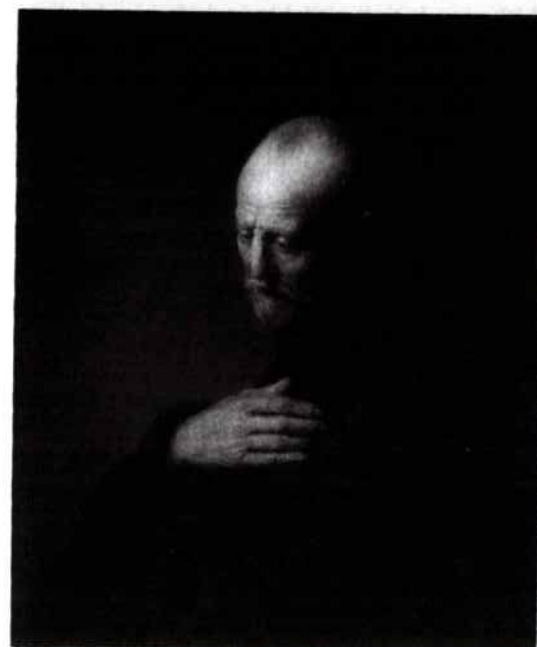


FIGURE 5.13



FIGURE 5.14



standing of smooth if we juxtapose it to rough. It is a physical fact that if you touch something rough and grainy, and then touch something smooth, the smooth will seem smoother. Opposites seem more intensely themselves when you think of them in terms of their uniqueness. In this observation lies the essential meaning of the word contrast: to stand against. By comparing the unlike, we sharpen the meaning of both. Contrast is a vital pathway to clarity of content in art and communication. Susanne Langer, in her essay, "The Dynamic Image,"\* says of this phenomenon: "A work of art is a composition of tensions and resolutions, balance and unbalance, rhythmic coherence, a precarious yet continuous unity. Life is a natural process of such tensions, balances, rhythms; it is these that we feel, in quietness or emotion, as the pulse of our own living." But the pull demonstrated by contrast between opposites must be handled as delicately as spice in cooking. The main purpose of a visual statement is expression, the conveying of ideas, information, and feelings; it must be seen in terms of expression to be understood most easily. Rudolf Arnheim has given the most creative interpretation to the interaction of thought and visual stimuli. In his essay, "Expression and Gestalt Theory," included in a broad collection of writings called *Psychology and the Visual Arts*, Arnheim defines expression as the "psychological counterpart of the dynamic processes which result in the organization of perceptual stimuli." In other words, the very same means the human organism uses to decode and organize and make sense out of visual information, all information for that matter, could serve most effectively for composing a message to be viewed by an audience. The process of human informational input, in its psychological and physiological ramifications, could serve as a model for informational output.

Whether at the level of expression that involves only the contrast of visual elements, or at the level of expression that involves conveying complex visual information, a visual communicator must recognize the efficient character of contrast, and its importance as a working tool that can be used and should be used in visual composition. Contrast is the sharpener of all meaning; it is the basic definer of ideas. There is so much richer an understanding of happiness when it is thought of juxtaposed with sadness; love with hate; affection with hostility; motivation with passivity; involvement with loneliness. Each purely conceptual polarity can be associated with and expressed by visual elements and techniques that can be associated with

\*In *Problems of Art*.

their meaning. Love, for instance, would be suggested by curves, circular shapes, warm colors, soft texture, similar proportions (5.15). Hate, as its polarity, would be reinforced by angles, straight shapes, hot colors, rough texture, dissimilar proportions (5.16). The elements are not absolutely opposite, but almost. Of all the visual techniques, it is contrast that is omnipresent in effective visual statements at all levels of the comprehensive structure of the message, conceptual and elemental. So it should be noted that contrast as an invaluable visual tool must constantly be referred to, from the generalized stage in visual composition right down to the specific character of each visual element chosen to articulate and express ideas visually.

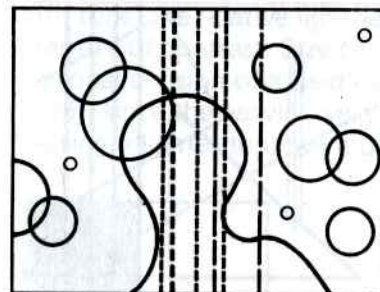


FIGURE 5.15

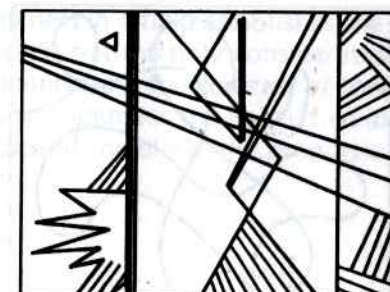


FIGURE 5.16

It is obviously more efficient to explain high by contrasting it with low, particularly in the use of visual stimuli (5.17). Proportion is vitally important in the manipulation of the field compositionally. To express precisely the emphasis of dissimilarity of visual clues, there-

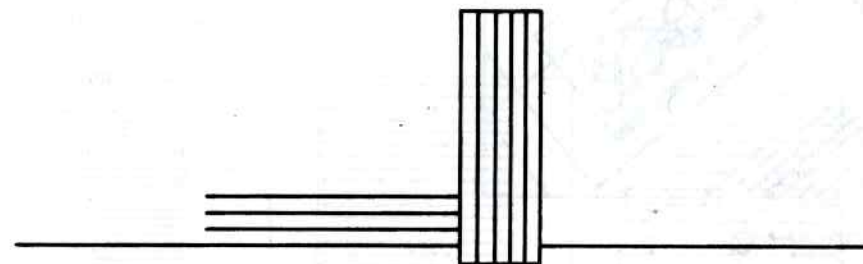


FIGURE 5.17

fore, the major point should have the largest proportion of space devoted to its display (5.18), at the least a two-thirds to one-third divi-



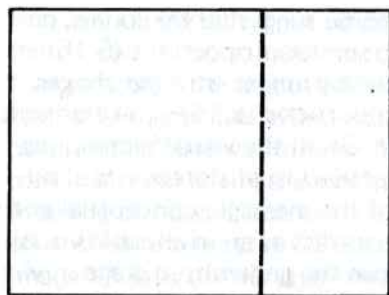


FIGURE 5.18

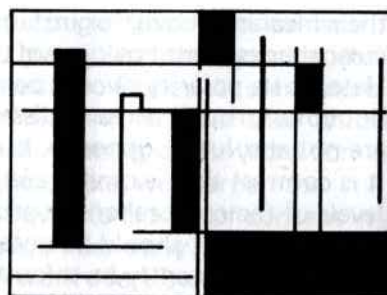


FIGURE 5.19

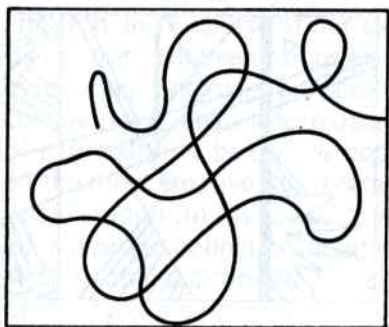


FIGURE 5.20

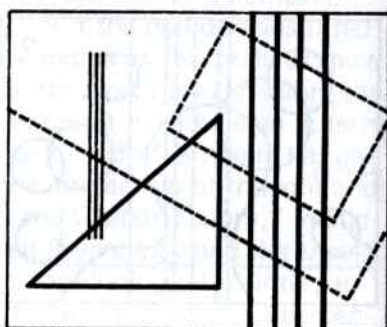


FIGURE 5.21

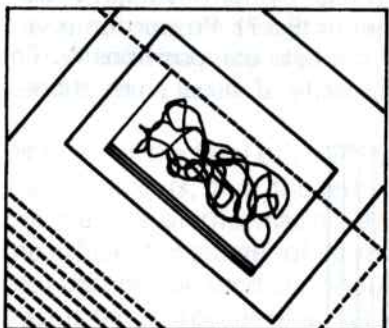


FIGURE 5.22

sion. This proportional division should increase the preciseness of intention in the composition (5.19). Whatever the effect intended, the primary information should have a disproportionate, large area of the field devoted to it. Proportion and scale depend on manipulation of size or space for effect, but, while this is a basic consideration in

the structure of contrast, it is not absolutely necessary. Other elemental forces count for a great deal in the final effect. Each visual element offers multiple possibilities in the producing of contrasting visual information. Line, for instance, can be formal or informal, and each carries strong informational clues. The looseness of informal line creates a feeling of investigation and unresolved probing (5.20), while the formal use of line connotes precision, plan, technics (5.21). Only in the juxtaposition of the two opposites can we create a contrasting composition (5.22) in which the basic character of each line treatment is accentuated.

### CONTRAST IN TONE

With tone, the relative lightness or darkness in a field establishes the intensity of contrast. Size or proportion is not the only consideration. An equal division of a field can still demonstrate the contrast of tone (5.23), since the heavier weight of black dominates the field. If an increasingly lighter tone were used in place of the black, the proportion



FIGURE 5.23

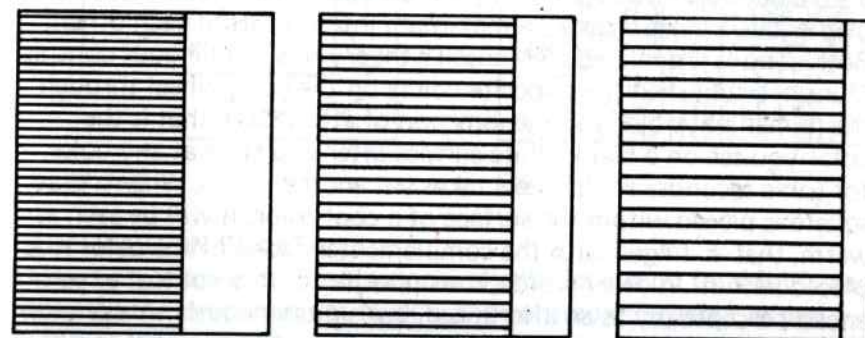


FIGURE 5.24



of the area covered by the deeper tone would have to be increased in order to sustain the effect of dominance and recessiveness that offers visual reinforcement to conceptual messages (5.24). Of course, tone is not usually distributed in a field in this regular and rigid way, and, yet, analysis of a visual composition can show whether there is a substantial enough division of tonal extremes toward an end of expressing contrast. Rembrandt went to great extremes to control his composition by using intense contrast, light against dark, dark against light, for the most compelling visual results in history.

### CONTRAST IN COLOR

Tone supersedes color in our negotiation of the environment, and, therefore, tone has far more importance than color in the establishment of contrast. Of the three dimensions of color (hue, tone, chroma), it is the tone that dominates. Johannes Itten established a structural approach to the study and use of color based on many contrasts, primarily emphasizing the light-dark opposition. Probably the most important contrast in color, beyond the tonal, is the cold-warm contrast, which divides the red-yellow dominated warm colors from the blue-green, cool colors. The quality of recessiveness in the blue-green range has been used to indicate distance, while the dominant quality of the red-yellow range has been used to express expansiveness. These qualities can affect spatial position since the color temperature can suggest proximity and distance. Itten cites some other color contrasts, among them, complementary and simultaneous. Each deals with the quality of color that can be used to sharpen a visual statement. Complementary contrast in color is the relative balance of cool and warm. According to the Munsell theory of color, the complementary is the exact opposite on the color wheel. In pigment form, the complementaries demonstrate two things: first, when mixed together, they produce a neutral, median tone of gray; second, when juxtaposed, complementary colors provoke a maximum intensity from each other. Both actions are related to Munsell's theory of simultaneous contrast. Munsell established the opposite colors on the color wheel through the human physiological phenomenon of afterimage, that is the color you see on a blank white surface after you stare at any color for some seconds. The process takes yet another form. When a gray square is placed within the surface of a cool color, it will be seen as warm, that is, tinged with the complementary tone of the color it is placed within. In other words, the opposite color is not just experienced perceptually as an afterimage, but simultaneously in a physiological process of neutralization that relates to the apparent pull to-

ward reduction of all visual stimuli to the most neutral and simplified form possible. We feed the complementary color into whatever color we are viewing. So, it would appear, not only are we experiencing a constant stimulus reducing effect in our pattern perception, but also, we are physiologically engaged in a color erasing process in our visual informational input in a never-ending pursuit of a middle tone of gray. Contrast is the key antidote to this action.

### CONTRAST IN SHAPE

The need of the whole system of human perception for leveling, for absolute balance, for visual closure, is the action against which contrast sets up a counteraction. In each instance of citing a basic visual element, we can quickly demonstrate the dynamics of contrast by constructing a compositional counterforce. Regular shape, simple and resolved, is dominated by the irregular, the unpredictable, in terms of attracting the attention of the viewer (5.25). Dissimilar textures intensify the unique quality of the other when they are juxtaposed (5.26). The same factors, of disproportionate and differentiated qualities juxtaposed, figure in the use of all the visual elements for the purpose of taking advantage of the value of contrast in the definition of visual meaning. The major function of the technique is sharpening through dramatic effect, but at the same time, it very successfully can refine the entire mood and sensation of a visual statement. Contrast must augment the designer's intentions.

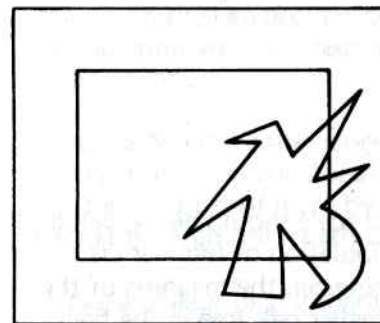


FIGURE 5.25

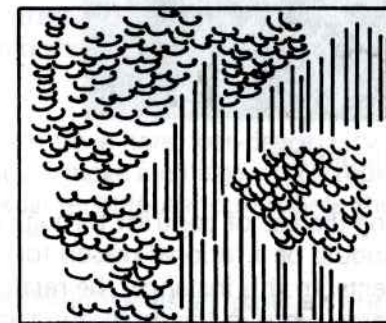


FIGURE 5.26

### CONTRAST IN SCALE

Distortion of scale, for instance, can shock the eye by forcefully manipulating the proportion of objects and contradicting what experience has led us to expect to see (5.27). The idea or message underly-



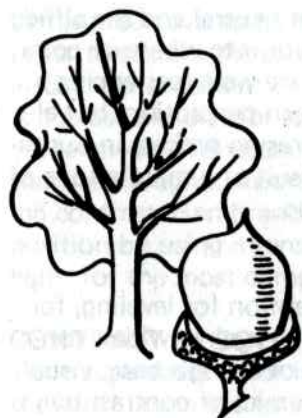


FIGURE 5.27



FIGURE 5.28

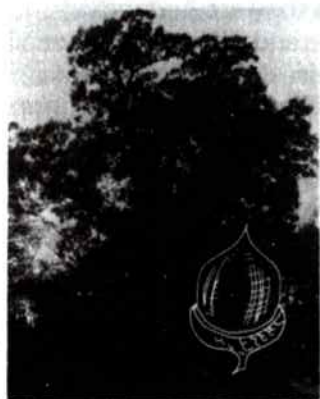


FIGURE 5.29

ing the use of contrast through distorted scale should be logical; there should be a rational reason for the manipulation of familiar visual objects. In this instance, the relationship between the meaning of the large acorn in the foreground and the smaller oak tree in the background, reverses the "great oaks from little acorns grow" idea visually, yet it dramatizes the importance of the acorn and, by doing so, articulates the basic meaning intended. As a visual technique, contrast can be further emphasized by juxtaposing different media. If the acorn is rendered in tone and the tree in line (5.28) or the tonal rendition is a photograph and the line drawing is looser and more interpretive (5.29), the contrast is increased through elemental visual cues from which we discern meaning.

Contrast, at the basic construction and decoding level, can be utilized through all the basic elements: line, tone, color, direction, shape, movement, and most particularly, proportion and scale. All are valuable forces in the ordering of visual input and output, emphasizing the key importance of contrast in the control of meaning. Any visual message combines the elements in a complex interaction. Many things are happening at once, and it is difficult to avoid confusion and ambiguity. If the final effect is to be coherent, the vague and the generalized must be modified by contrast toward the precise and the specific state of the concrete in a decision-making process of design. Seeing moves toward organization of data, reaching from the primary sensations, the expression and comprehension of simple ideas through increasing complexity to the abstract. Visual information has the same evolving character, but at any point in the hierarchy, it must be disciplined by the designer's intention to communicate. Whether in the case of an arrow cut into a tree to mark a pathway through the woods, or an imposing cathedral reaching its towers toward the sky, the organization of visual elements must respond to the purpose of the visual statement, the form must follow the function. In this pursuit, contrast is the bridge between the defining and comprehension of visual ideas, not in the verbal sense of definition, but in the visual sense of making ideas and sensations and images more visible.

### EXERCISES

1. Make a photograph or find examples of both (1) a balanced, harmonious and (2) an asymmetrical, contrasting visual statement. Analyze and compare the effect of each and their ability to convey information or mood.
2. Choose two opposite conceptual ideas (love-hate, war-peace, city-country, organized-confused). In a square make a collage which represents the contrast of ideas by using visual techniques to reinforce the meaning through the visuals used.
3. Make a collage or photograph in which unlike visual materials are juxtaposed toward a heightened or sharpened effect of the message.
4. Look for an example of design or graphics where surprise in the juxtaposition of unexpected visual information dramatizes the artist's underlying intention.